

From TEM Micrograph to UV-Vis Speciation: an Automated Sizing and Optical-Physics Pipeline for Gold Nanoparticle Colloids

Mid-project progress report

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Why this matters (plain-language summary for life-science readers)

The problem, in one sentence. Gold nanoparticles are a workhorse of bionanotechnology — contrast agents, biosensors, drug-delivery cargo — and their usefulness depends on two numbers that are surprisingly hard to measure routinely: how big are the particles, and are they clumping together? Both are answered today by an expert spending hours at an electron microscope, hand-clicking a few hundred particles per sample. This project replaces that bottleneck with two automated tools and connects them, so a single cheap optical scan plus one microscope image yields size, size spread, and the fraction of gold that is aggregated.

Where it helps a biology lab. This work was built directly on data from the Dragnea lab (Indiana University), which co-assembles virus-like particles (VLPs) — coat-protein shells of the brome mosaic virus — around gold-nanoparticle cargo, both as a route to engineered delivery vehicles and as a model system for how viruses package “fragmented” (multi-piece) cargo such as segmented genomes, multiple siRNAs, or several drug molecules [25,26]. The nanoparticle plays the cargo role precisely because gold is visible: it gives the electron-microscope contrast that lets the assembly be read particle-by-particle. Two features of that research make our tools a direct fit:

- The whole biology is gated by nanoparticle size. Encapsulation is strongly size-selective — particles below a few nm are excluded, larger ones are preferentially packaged, and the number of nanoparticles per shell falls with their diameter [25,26]. The key evidence is a shift between two size distributions — the free nanoparticle stock versus the population actually encapsulated — and the lab establishes it by hand-counting 300–500 particles per TEM sample, for both populations. Stage one below automates exactly that count: it turns a half-day of clicking into minutes, measures several-fold more particles, and gives the less outlier-biased distributions that decide whether a size-selection signal is even resolvable. Sharper size distributions in, sharper thermodynamic conclusions out.
- Assembly is a speciation equilibrium — the same problem this project models. The lab’s own theory describes co-assembly as a mass-action balance between free nanoparticles, nanoparticle–protein complexes, and capsids holding a variable number of packed cores [26] — and inside a multi-core VLP those cores sit as adjacent linear or clustered arrays against the shell wall [25], currently resolved only by cryo-electron tomography. That is precisely the situation our optical model was built for: gold particles packed at contact couple plasmonically, and the strength of that coupling reports how many are clustered and how tightly — a non-destructive, in-solution, high-throughput readout of assembly/cluster state that could

complement low-throughput cryo-EM. The coupling signal is quantitative at the $\geq \sim 10$ nm sizes typical of single-core AuNP–VLP work and of routine colloid QC; extending it to the few-nm fragmented cargo is a natural next step rather than a solved capability today.

In short, the two measurements this class of nanoparticle-cargo and VLP research needs on every batch are throughput — automated sizing that scales to thousands of particles across both free and assembled populations — and aggregate/cluster inference — a quantitative clustering readout from a one-minute optical scan, grounded in the same equilibrium picture the assembly itself obeys. Both now come out of one connected pipeline, with stated error bars and honest failure flags, and the absolute-extinction concentration measurement the lab already runs [19] is the anchor that ties them together. The technical report follows.

1. Summary

We have built an end-to-end pipeline that starts at a raw electron-microscope image and ends at a quantitative statement about a colloid's state in solution. Stage one measures every gold nanoparticle in a TEM micrograph automatically; stage two feeds that size distribution into a first-principles optical model that predicts the UV-Vis spectrum with zero fitted parameters; stage three reads the deviation between prediction and measurement for what it physically is — line-width chemistry and aggregation. Four results define where we stand:

1. Automated TEM sizing is validated. On all eight of the lab's samples the automated mean diameter agrees with hand-picked measurements to within $\pm 6\%$ (mostly $\pm 3\%$) while measuring $1.6\text{--}6.4\times$ more particles; a foundation-model segmentation tier correctly separates touching particles inside aggregates that defeat classical thresholding.
2. TEM size alone predicts the spectrum of a clean sol. For well-dispersed samples the predicted plasmon peak lands within a few nanometers of the measurement with nothing fitted — and the automated sizes drive the physics exactly as well as manual counting (peaks within 1 nm). Where the prediction fails (the two aggregated samples), the failure is the aggregation signature the project exists to quantify.
3. Line width is a per-synthesis-route material property, and with it calibrated, speciation becomes quantitative. A two-parameter damping law calibrated on verified-clean monomers reduces model-data mismatch $\sim 2.5\times$ per route; CTAC-grown and seeded-growth H_2O_2 /citrate particles land on opposite sides of textbook gold ($s = 0.05$ vs 1.75) — plasmon width doubles as a crystallinity/interface probe. With the calibration in place, a physically-pure 12 nm control that previously read a spurious 13% “aggregation” now reads 0%, while the anomalous 20 nm batch keeps a real $\sim \frac{1}{3}$ of its gold in small clusters (31–38% across model choices) — answering the question the PIs posed about that batch. The 40/60 nm batches fit at $\approx 46\%$ and $\approx 81\%$; the flagged “55 nm” batch is beyond the small-cluster regime entirely.
4. The temperature series shows no thermal monomer $\rightleftharpoons$ dimer signal — but two real, unexplained observables remain. After correcting a normalization pitfall (evaporation, $\sim 5.6\%$ over the run) and subtracting measured blanks, the 13 nm citrate/TEG heating/cooling series shows an isosbestic-like crossing that drifts 544 $\rightarrow$ 568 nm (a fixed two-state equilibrium cannot produce this; an evolving cluster geometry can [21]), and a flat, reversible scattering pedestal ($\sim 15\%$ of peak) that survives every instrumental exclusion we could test. A DLS ramp or filtration control is the decisive lab experiment.

2. Motivation

Manual TEM sizing is slow, samples relatively few particles, and its histogram widths are inflated by whichever outliers a person happens to click. Worse, two batches that look identical by hand-measured histogram can behave completely differently in solution — because one is aggregating, and a dried-grid histogram cannot see it. Conversely, an honest UV-Vis “size” readout is confounded by aggregation and by chemistry-dependent line width. The goal is an instrument-grade chain: an unbiased, high-n per-particle measurement that reproduces trusted hand measurements, plus an inverse optical model that, given a routine UV-Vis spectrum, reports nominal size, size spread, and speciation — the fraction of gold bound in small clusters — with stated uncertainties and honest failure flags. The division of labor that emerged from this work: TEM pins the size; the spectrum then yields the aggregated fraction as a stable linear readout.

3. Stage one — automated TEM sizing (image → per-particle table)

The tool reads native Gatan dm3 micrographs directly (RosettaScilO/HyperSpy [7]); pixel calibration comes from file metadata, so no burned-in scale bar is needed and the calibration is exact. The present dataset is 47 frames across 8 samples (Oct 2024: GNP12/20/40/60; Nov 2024: seed-11/23/39/55). Segmentation has two interchangeable tiers that emit the identical per-particle table (diameter, area, circularity, aspect ratio, shape class, touching-group flag):

- Tier 1 — classical (no GPU, ~0.5 s/image): background flattening → Otsu threshold [1] → distance-transform watershed [2] with a rod-aware neck-merge, built on scikit-image [3]. Accurate on well-dispersed fields; merges particles inside dense aggregates, as expected.
- Tier 2 — foundation model (~6 s/image on a laptop GPU): NP-SAM [6], a nanoparticle-EM-tuned wrapper around the fast Segment-Anything variant FastSAM [5] (SAM [4]). It recognizes particles by learned shape/edge features and separates touching particles inside clusters; a physics-based “ghost” filter removes background false positives.
- Human-in-the-loop QC: segmentation is never blindly trusted — a one-click reviewer lets a person drop any detection, with statistics recomputed live; decisions are stored separately and never overwrite the machine output.

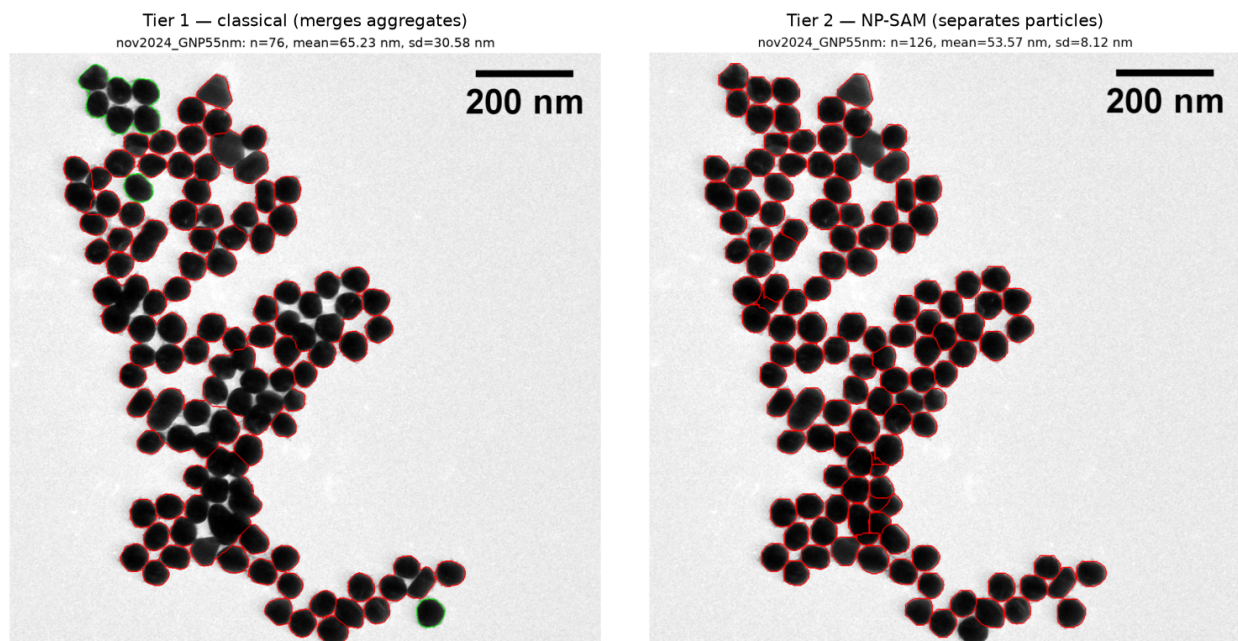


Figure 1: Same aggregated sample (Nov “55 nm”). Left: tier-1 classical segmentation merges touching particles into oversized blobs (mean 65 nm, sd 31 vs a true 54.6 ± 4.9 nm). Right: tier-2 (NP-SAM/FastSAM) separates the individual particles (mean 54 nm, sd 8), recovering the true size.

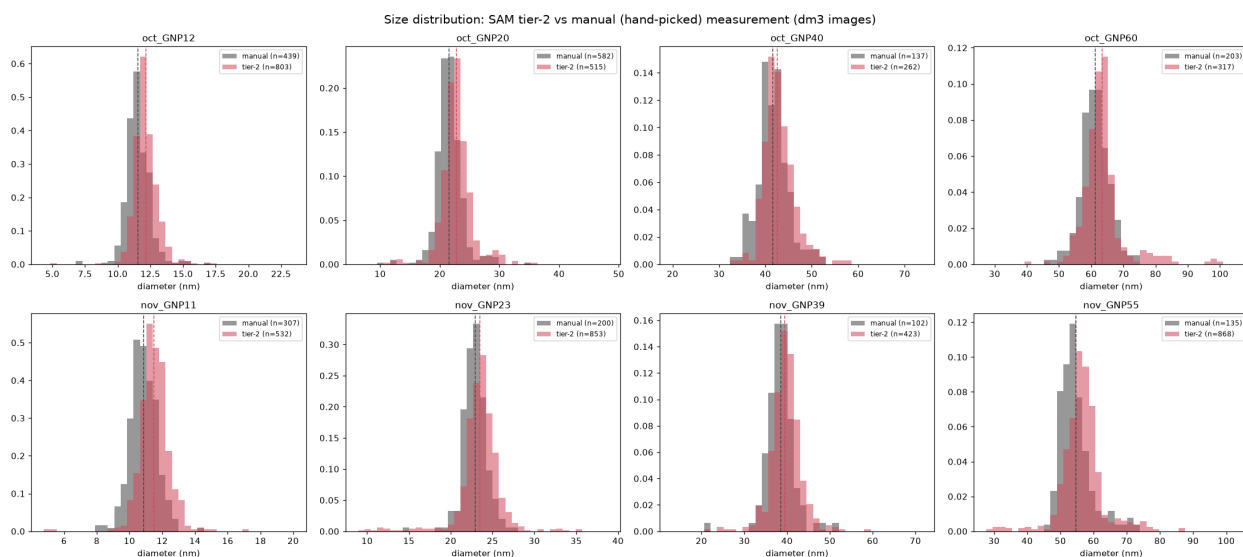


Figure 2: Size distributions: automated tier-2 (red) vs hand-picked (grey), per sample, from the dm3 micrographs. Dashed lines are the means.

Validation against hand measurement. Both tiers ran over all 47 frames; aggregated distributions were compared to the lab’s hand-picked per-particle measurements (Figures 1–2):

Sample	Manual (n / mean $\pm$ sd)	Tier-1 (n / mean, Δ)	Tier-2 (n / mean, Δ)
oct GNP12	439 / 11.6 $\pm$ 0.9	846 / 11.6 (+0%)	803 / 12.2 (+5%)
oct GNP20	582 / 21.6 $\pm$ 2.2	937 / 21.9 (+1%)	515 / 22.8 (+6%)
oct GNP40	137 / 41.6 $\pm$ 3.3	137 / 52.2 (+25%)	262 / 42.5 (+2%)
oct GNP60	203 / 61.3 $\pm$ 4.5	101 / 96.1 (+57%)	317 / 63.3 (+3%)
nov seed-11	307 / 10.9 $\pm$ 0.8	566 / 11.4 (+5%)	532 / 11.5 (+6%)
nov GNP23	200 / 22.9 $\pm$ 1.5	711 / 24.2 (+6%)	853 / 23.5 (+2%)
nov GNP39	102 / 38.5 $\pm$ 3.4	326 / 42.8 (+11%)	423 / 39.4 (+2%)
nov GNP55	135 / 54.6 $\pm$ 4.9	507 / 68.5 (+25%)	868 / 54.7 (+0%)

Tier-2 recovers the hand-picked mean within $\pm 6\%$ on every sample. The aggregated samples that defeat tier-1 (+11...+57%) all drop to +0...+3% with tier-2, while the automated extraction measures 1.6–6.4 $\times$ more particles (GNP55: 868 vs 135). The residual $\sim +5\%$ on the small samples is a known, size-dependent mask-boundary effect (SAM outlines run slightly generous) with a calibration in progress.

4. Stage two — closing the loop: zero-parameter spectrum prediction

Each automated size distribution feeds single-particle Mie theory [8] with Johnson-Christy gold constants [9] in water — nothing fitted; both data and model normalized to their own plasmon peak (a pure shape test):

Sample	Measured peak (nm)	SAM-predicted (Δ)	Manual-predicted (Δ)
oct GNP12	521	523 (+2)	523 (+2)
oct GNP20	523	524 (+1)	524 (+1)
oct GNP40	528	528 (+0)	528 (+0)
oct GNP60	545	537 (−8)	536 (−9)
nov seed-11	518	523 (+5)	523 (+5)
nov GNP23	520	524 (+4)	524 (+4)
nov GNP39	525	527 (+2)	527 (+2)

Sample	Measured peak (nm)	SAM-predicted (Δ)	Manual-predicted (Δ)
nov GNP55	580	534 (−46)	533 (−47)

Three observations: automated and hand-picked predictions are indistinguishable (within 1 nm) — the automated table drives the physics exactly as well as manual measurement; for well-dispersed samples the measured peak is predicted within a few nanometers from TEM size alone; and GNP60 and especially GNP55 fail by design — spectra far redder and broader than single-particle physics allows. That red-shift is the plasmonic-coupling signature of aggregation, and quantifying it stages three and four below.

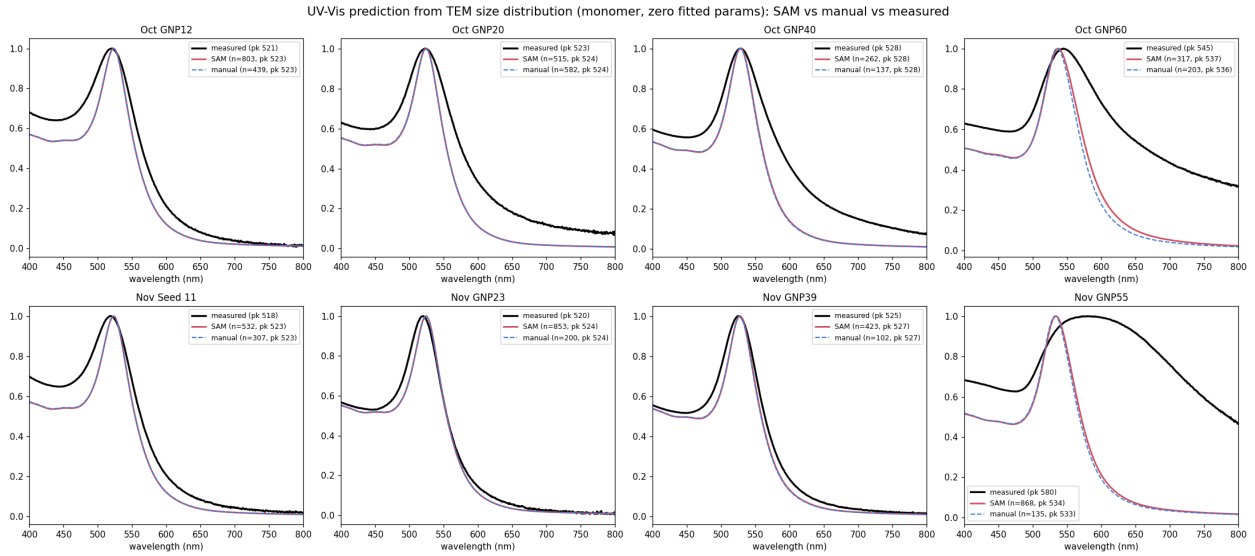


Figure 3: UV-Vis predicted from the TEM size distribution (SAM red, manual blue) vs measured (black), normalized to peak. Well-dispersed samples match within a few nm; GNP55/60 show the aggregation red-shift the monomer model cannot reproduce.

5. The physics engine

The optical forward model has three layers, all validated against internal and external references:

- Optics. Single-sphere Mie theory [8] with size- and temperature-corrected gold permittivity; small clusters via two selectable backends: a fast coupled-dipole approximation (CDA), and an exact multipole T-matrix solver (`trems` [17]; equivalent to Mackowski-Mishchenko MSTM [18]). The exact backend reproduces Mie for a single sphere to machine precision and shows the CDA under-couples at near-contact — $1.6\times/2.8\times$ less dimer/trimer red-tail at 1 nm gap — so CDA is used for exploration and the T-matrix for quantitative work, cached on parameter grids for fitting speed.
- Materials. Gold $\epsilon(\lambda)$ from Johnson & Christy [9] (selected against Etchegoin [10], Brendel-Bormann, and Yakubovsky [11] alternatives; a nine-sample re-anchoring on verified-clean monomers measured a $+2.7 \pm 1.1$ nm red bias, now applied as a fixed offset), Drude parameters from Olmon [12]; surface damping $\gamma_S = A \cdot \hbar v_F / R$ [13,14]; bulk damping $\gamma(T)$ bracketed

between Holstein electron-phonon theory [15] and Reddy's measured film values [16]; water $n(T)$ and $k(\lambda)$ from standard tables [23,24]; solvent density/evaporation corrections in the data layer (§9).

- Populations & inversion. Monomer $\rightleftharpoons$ dimer $\rightleftharpoons$ trimer equilibrium with van 't Hoff $K(T)$; inversion by non-negative least squares over species basis spectra (linear, given size) inside a nonlinear fit over size/spread, with profile-likelihood identifiability checks; a global multi-temperature fit ties a whole T-series to one sample with shared thermodynamics.

6. What one spectrum can — and cannot — determine

A methodical identifiability study (synthetic recovery tests plus validation on five CTAC monomer samples with paired TEM) established limits that shape everything downstream, consistent with the sizing literature [19,20]:

- Robust: the aggregated (non-monomer) gold fraction. With size pinned, it is a stable linear readout; on clean monomer samples the pipeline correctly returns zero (negative control passed).
- Weak: mean size from normalized band shape below ~ 35 nm (a 23 nm sample fits equally well at 8 nm); the Haiss A_{spr}/A_{450} ratio [19] separates 8 from 30 nm but saturates over 29–42 nm. Size must come from TEM, absolute extinction, or a temperature series — not band shape alone. This is exactly why stage one (automated TEM) is the right partner instrument.
- Not identifiable: dimer vs trimer vs tetramer (collinear red tails — only the total aggregated fraction is quotable); blind bimodality (a two-size mixture is spectrally indistinguishable from one broad population); and speciation at ~ 8 nm (profile bounds explode).
- Conditional: at ~ 13 nm the monomer/dimer separation depends on the assumed interparticle gap (identifiable at 1 nm contact, degenerate at $\geq 2\text{--}3$ nm) — gap priors from ligand length matter.

7. Line width is chemistry: per-route damping calibration

The single largest systematic in all fits was the monomer line width. We model the effective Drude damping as $\gamma_{\text{eff}} = s \cdot \gamma_{\text{bulk}} + A \cdot \hbar v_F / R$ — s scales the tabulated (evaporated-film) bulk ϵ_2 , absorbing crystallinity differences; A is the surface/interface term [13,14]. One (s , A) pair is fitted per synthesis route, jointly across that route's verified-clean monomer sizes (the $1/R$ law carries the size dependence):

route	calibration set	(s , A_{surf})	mean shape RMS
CTAC (7.8–42 nm)	5 samples, per-particle TEM	(0.05, 0.6)	4.6% $\rightarrow$ 2.5%
seeded-growth citrate/H ₂ O ₂ (11–39 nm)	4 clean samples	(1.75, 1.0)	7.7% $\rightarrow$ 3.0%

The two routes bracket textbook gold from opposite sides: CTAC particles are less damped than Johnson-Christy's evaporated films (single-crystal-like; J&C ϵ_2 effectively 15–35% too high

for them), while seeded-growth H_2O_2 overgrowth is substantially more damped (defect-rich / multiply twinned), with an interface term at the top of the literature range [14]. A frozen calibration transferred across routes makes fits worse — the pair is a route property, not a universal constant. Two practical consequences: the plasmon width can serve as a fast crystallinity/interface probe; and any speciation fit must use its route’s calibration or it will convert line-width error into fictitious aggregation (§8).

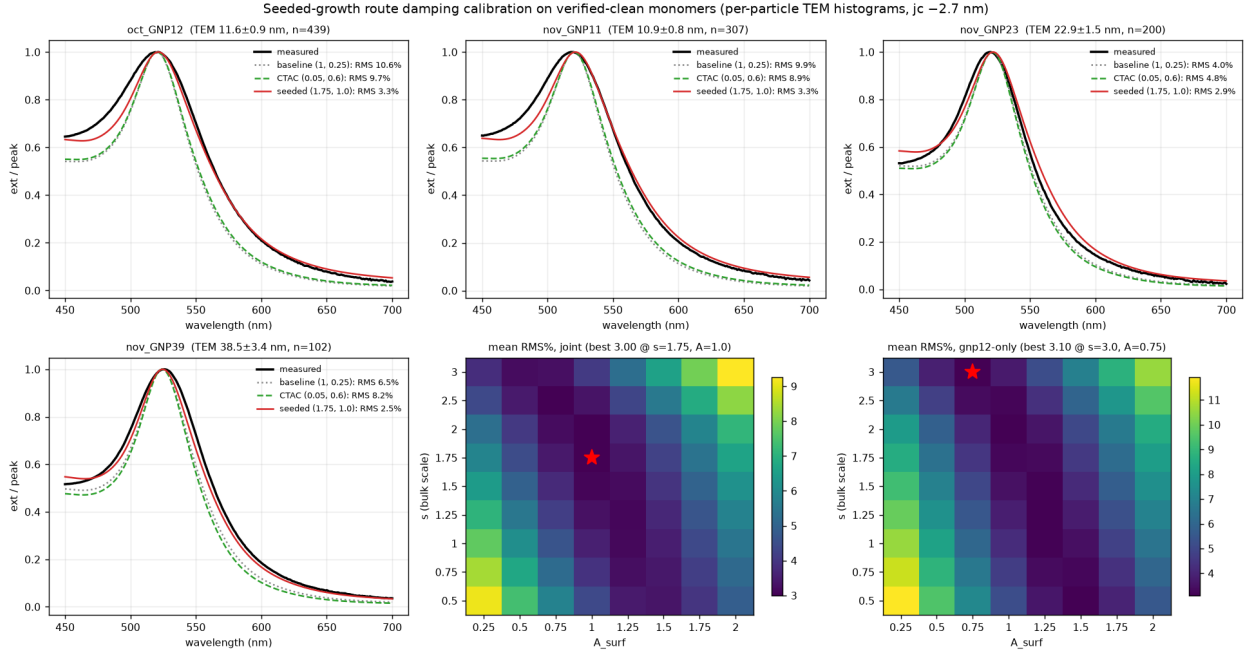


Figure 4: Seeded-growth route calibration on the four verified-clean monomers (per-particle TEM histograms pinned; $j_c -2.7$ nm offset). Measured spectra (black) vs the uncalibrated model, the (wrong-route) CTAC calibration, and the seeded-growth calibration (red). Right panels: RMS landscape over (s, A) — a diagonal valley (the constrained combination is the total damping) with an interior joint optimum at $(1.75, 1.0)$.

8. Stage three — TEM-pinned speciation of the seeded-growth series

With sizes pinned by stage one and the route-calibrated monomer, fitting each spectrum as a non-negative monomer + dimer + trimer + tetramer mixture is a clean linear problem. The first-cut fit (uncalibrated line width) had assigned 13% “aggregation” to the physically-pure 12 nm control — the tell of an under-damped monomer model padding width with fake aggregate. After calibration:

sample	first-cut agg	calibrated CDA	calibrated exact T-matrix (gap range)
oct GNP12 (11.6 nm, pure control)	13%	0%	—
oct GNP20 (22.8 nm)	41%	45%	31% (31–38%)
oct GNP40 (42.5 nm)	40%	38%	46% (41–46%)
oct GNP60 (63.3 nm)	68%	67%	81% (75–81%)

sample	first-cut agg	calibrated CDA	calibrated exact T-matrix (gap range)
nov series (11/23/39 nm, clean)	6/0/7%	0/0/7%	—
nov GNP55 (54.7 nm, flagged)	100%	100%	100%, structured misfit

The diagnostic splits exactly as hoped: the pure-monomer control collapses to zero (so the calibration is right), while GNP20 does not — its residual is red-asymmetric, which damping cannot and should not absorb. The Oct-series aggregation is real, and GNP20 carries roughly a third of its gold in small clusters, robust across optics backends (CDA vs exact) and the full interparticle-gap grid. Nothing was calibrated away: the clean Nov samples stay clean with improved residuals.

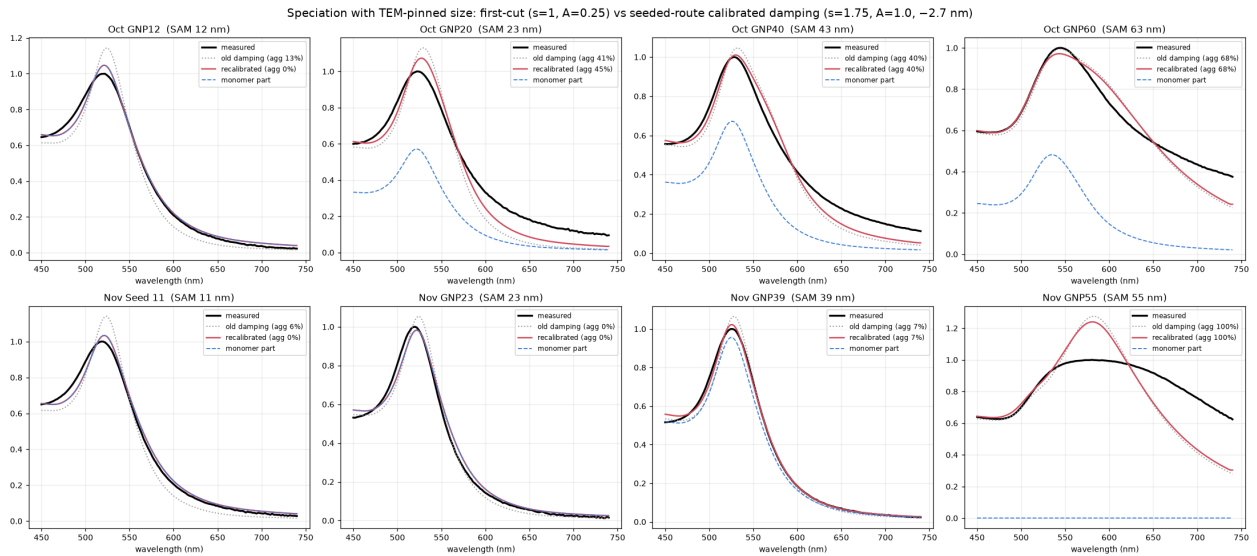


Figure 5: Speciation with TEM-pinned size, before (dotted) and after (red) the route damping calibration. The pure 12 nm control (top left) drops from 13% to 0% aggregated; GNP20's misfit persists and is specifically red-sided — real coupling, not line width.

Three honest caveats. (i) The exact-optics fractions are gap-conditional ranges, quoted as such. (ii) The intuition “CDA under-couples, so its fractions are upper bounds” proved wrong at large sizes — exact chain modes concentrate red intensity in shaped bands rather than uniformly boosting the tail, and the exact fractions for GNP40/60 came out above CDA's; we therefore quote the CDA→exact spread as the backend systematic. (iii) The dimer/trimer/tetramer split within the aggregate is not identifiable (collinear red tails; the fitter parks weight at corners of the degenerate family) — only the total aggregated fraction is reported. GNP55 remains honestly unfittable (its smooth 580 nm plateau defeats any single-gap small-cluster basis): that sample is in a larger-floc regime, matching the experimenter's own “redo” flag.

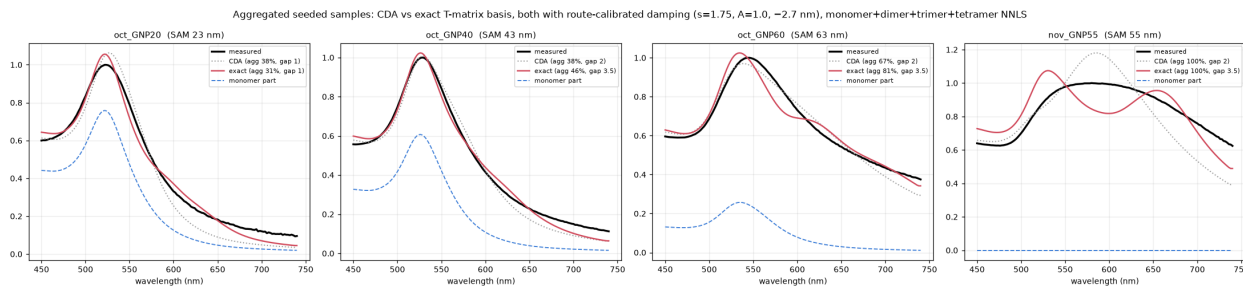


Figure 6: CDA vs exact T-matrix fits of the aggregated samples, both with calibrated damping. The exact basis halves the GNP20 residual; GNP60’s fitted chain modes show a ~620 nm bump the smooth data lack — the signature of a single-motif basis, motivating the ensemble-averaged (Monte-Carlo geometry) basis planned next.

9. The temperature series: no thermal speciation — and what remains

The original motivation was a temperature-dependent “isosbestic point” in a 13 nm citrate/TEG sol, read as evidence of a two-state monomer $\rightleftharpoons$ dimer equilibrium. Careful data handling reversed both readings:

- Normalization pitfall (methodological result worth publishing on its own). Heating/cooling branches at matched temperature differ by a wavelength-flat +5.6% — the cell lost ~5.6% of its water to evaporation over the run, overwhelming the -2.4% thermal-expansion correction [22] and masquerading as a growing flat pedestal. We built a branch-ratio diagnostic that decomposes any dual-branch series into concentration drift vs real spectral change, and an evaporation-aware normalization calibrated from the data themselves (one parameter; reproduces a hand-computed concentration table to $\leq 0.65\%$). Per-temperature water blanks are subtracted before any normalization.
- No robust thermal speciation. With gold $\epsilon(T)$ [15,16], water $n(T)$ [23], and surface damping in the basis, the fitted ΔH_2 is consistent with zero on both branches and the aggregated fraction is nearly T-flat (~0.15–0.21, opposite weak trends on heating vs cooling). What temperature actually moves is the gold and solvent dielectrics — bulk gold measurably does this at exactly the magnitude used [16] — plus the pedestal below.
- The crossing drifts. The difference-spectra crossing walks 544→568 nm over 30→75 °C (cooling ~flat at 546–554 nm). A fixed-geometry two-state system produces a stationary crossing (our model: drift <1 nm even with full $\epsilon(T)$); a drifting one indicates evolving cluster geometry — precisely the behavior Cardellini et al. recently established, with λ_{iso} tracking interparticle spacing [21]. The historical “575 nm isosbestic” was an artifact of the deprecated normalization.
- The pedestal is real but its origin is not decided by fitting. A near-flat scattering floor (~15% of peak) survives blank subtraction and every exclusion we could construct: its exponent is decisively not Rayleigh ($n \approx 0$ –0.9, not 4 — small-aggregate scattering ruled out), it is not a permittivity artifact, not stray light drift (blanks are T-flat), not thermal-optical cell effects, and not surface nanobubbles (wrong wetting, size, and T-dependence). It grows reversibly by ~10–16% over 15→75 °C. Both μm -scale flocs and a static stray-light floor fit the shape; DLS across the ramp or a filtration control is the decisive experiment — we recommend it as the next lab action.

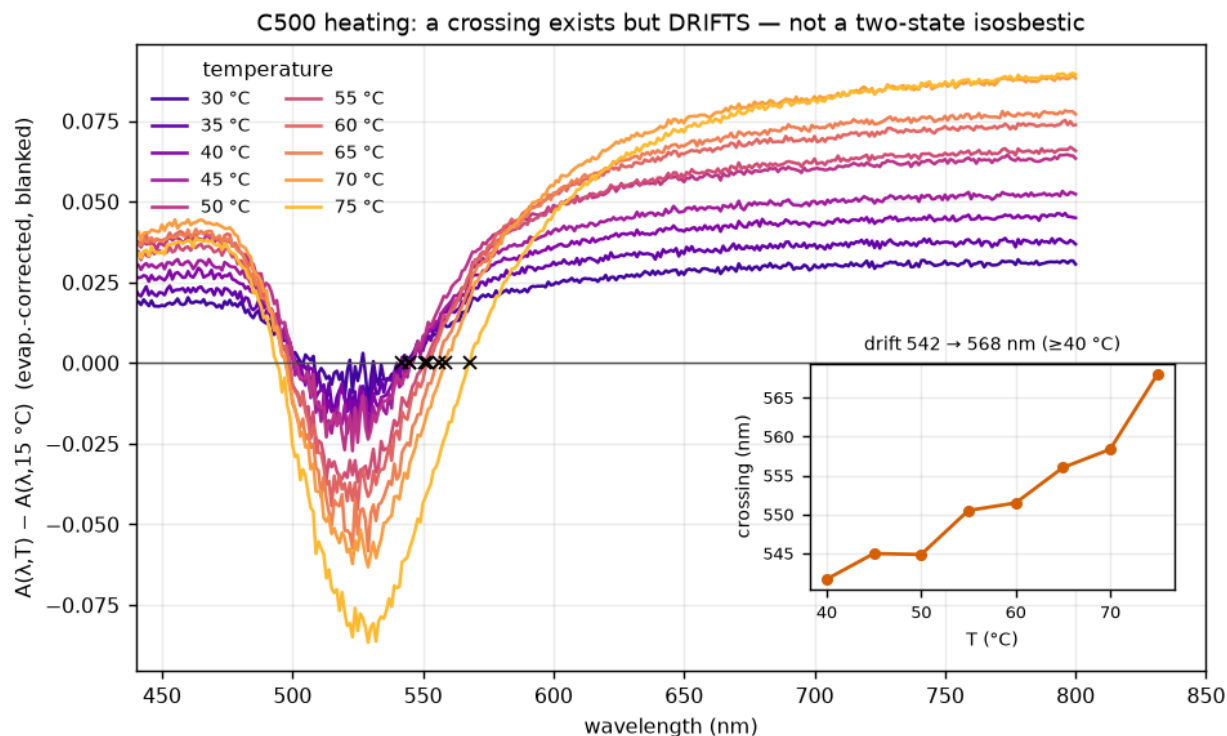


Figure 7: Data-level difference spectra (blank-subtracted, evaporation normalization): the crossing near 545–570 nm drifts with temperature on heating — incompatible with a fixed two-state equilibrium, consistent with evolving cluster geometry [21].

10. Status and next steps

Working now: dm3 input with exact calibration; two-tier segmentation validated against hand measurement on the lab's own images; mandatory QC overlays with one-click human review; per-particle output tables; a validated three-layer optical model (CDA + exact T-matrix, cached for fitting); route-calibrated damping for two chemistries; TEM-pinned quantitative speciation with stated identifiability limits; dual-branch normalization diagnostics; global multi-temperature fitting. The two repositories (tem-particle-metrics, aunp-speciation) share a clean data contract: the per-particle CSV table.

Next, in order of leverage:

1. Ensemble (Monte-Carlo geometry) cluster basis. Replace single-motif, single-gap cluster columns with averages over sampled configurations (N, motif, gap distribution) computed with the exact T-matrix. This addresses the two visible artifacts of the current basis (GNP60's spurious chain-mode bump; GNP55's unfittable plateau) and makes the non-identifiability of the N-split explicit.
2. Calibrate out the $\sim +5\%$ tier-2 size offset (mask-boundary effect), so absolute (un-normalized) extinction can be compared, not just peak shape; then use absolute extinction plus the A_{spr}/A_{450} ratio [19] to break the remaining damping-size-polydispersity degeneracy.
3. Lab: DLS ramp / filtration control on the temperature-series sample to settle the pedestal's origin (flocs vs instrument floor).

4. Adoption & packaging: the batch segmenter plus lightweight reviewer on a shared lab machine; per-route damping calibration as a standard step (one clean monomer series per chemistry); the branch-ratio drift diagnostic on every dual-branch dataset.
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